5. ADALINE (Adaptive Linear Neuron) – 1960

(a) Theory and Learning Algorithm

The Adaptive Linear Neuron (ADALINE) was introduced by Bernard Widrow and Ted Hoff in 1960. Unlike the perceptron, which uses a hard threshold activation function, ADALINE employs a linear activation function and minimizes a continuous error measure using gradient descent.

Model Formulation

Given an input vector $x \in \mathbb{R}^d$, ADALINE computes:

$$y = w^T x + b$$

where:

- $w \in \mathbb{R}^d$ is the weight vector,
- $b \in \mathbb{R}$ is the bias,
- y is the linear output.

For binary classification, the predicted class is obtained using:

$$\hat{t} = \text{sign}(y)$$

Loss Function

Unlike the perceptron, ADALINE minimizes the **Mean Squared Error (MSE)**:

$$E(w, b) = \frac{1}{2} \sum_{i=1}^{n} (t_i - y_i)^2$$

where

$$y_i = w^T x_i + b$$

and t_i is the target label.

Learning Rule (Widrow-Hoff Rule)

Using gradient descent, the updates are derived as follows:

$$\frac{\partial E}{\partial w} = -\sum_{i=1}^{n} (t_i - y_i) x_i$$

$$\frac{\partial E}{\partial b} = -\sum_{i=1}^{n} (t_i - y_i)$$

Using gradient descent, the update rule (Widrow-Hoff rule) becomes:

$$w \leftarrow w + \eta(t_i - y_i)x_i$$

$$b \leftarrow b + \eta(t_i - y_i)$$

where η is the learning rate.

(b) Convergence Guarantee and Independence from Data Separability

Convexity of the Objective

The MSE objective can be written in matrix form:

$$E(w) = \frac{1}{2}\|Xw - t\|^2$$

where:

- $X \in \mathbb{R}^{n\times d}$ is the data matrix,
- $t \in \mathbb{R}^n$ is the target vector.

Expanding the expression:

$$E(w) = \frac{1}{2}w^T X^T X w - w^T X^T t + \frac{1}{2}t^T t$$

The Hessian matrix of this objective is:

$$\nabla^2 E(w) = X^T X$$

Since $X^T X$ is positive semidefinite, the error function is convex. If $X^T X$ is full rank, it becomes positive definite, implying strict convexity.

Global Least-Squares Solution

Setting the gradient to zero:

$$\nabla E(w) = X^T(Xw - t) = 0$$

yields the normal equation:

$$X^T X w^* = X^T t$$

Thus,

$$w^* = (X^T X)^{-1} X^T t$$

which corresponds to the ordinary least-squares solution.

Because the objective function is convex, gradient descent with a sufficiently small learning rate converges to this unique global minimum.

Independence from Data Separability

Unlike the perceptron, ADALINE does not require linearly separable data.

The perceptron minimizes classification error, which is a non-convex objective and guarantees convergence only when the data is linearly separable.

In contrast, ADALINE minimizes squared error, which:

- is a convex objective function,
- always has a well-defined minimum,
- provides a least-squares solution even when perfect classification is impossible.

Therefore, ADALINE can still learn meaningful weight parameters even when the dataset is not linearly separable.